



# Rapid hypersonic sonic boom prediction using line-distributed energy impulse formulations with and without lift effects

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**Due to the growing interest in the design of low-boom hypersonic aircrafts, there is an urgent need for a theoretical sonic-boom prediction tool that is much cheaper than flight/shock-tunnel experiments and much faster than computational fluid dynamics (CFD). In this paper, we develop a novel hypersonic sonic boom theory that addresses this need. Specifically, a semi-analytic prediction tool based on a line-distributed energy impulse formulation, leveraging Newtonian aerodynamics for force distribution calculation, is developed for scenarios with and without lift. Since the process of solving detailed governing equations is circumvented using such a method, the algorithm was found to be much faster than CFD, with turnaround times in the order of a few minutes. The formulation was validated against CFD results with an error  $\sim 10\%$  by various scenarios with and without lift. Geometries used for validation include a sphere, a nosed cone, a nosed elliptic cone (the first sector of HIFiRE-5), and a boost-glide vehicle (BGV). The lift (or lack thereof) is achieved by varying the angle of attack. The formulation is also validated against wave propagation data (with CFD data as the input) for the middle and far fields, within acceptable errors. Azimuthal angle dependence (i.e., sonic boom carpet effect) and stratified atmosphere effect are also discussed.**

## I. Introduction

The development of advanced thermal protection systems together with optimized vehicle design has pushed the current limits of the maximum achievable flight Mach numbers to the hypersonic regime. However, another important design aspect in the pursuit of realizing hypersonic flight is the reduction of sonic boom signatures (i.e., over-pressure  $\Delta p = p - p_\infty$  signature). Hypersonic flight can result in prohibitively high levels of noise pollution [1] that needs to be mitigated in addition to satisfying the other design needs. Low-boom signature designs can also lower the associated energy costs thereby enhancing propulsion efficiency. The realization of these goals hence requires the precise and accurate prediction of sonic boom signatures for a given vehicle design. The coordinate system to describe the sonic boom problem is provided in Fig. 1

Historically, in the supersonic regime (which is the traditionally discussed regime in sonic boom research), the limitations in computational power led to the development of a time-saving theoretical strategy called the “F-function” method by Whitham [2] from linearized supersonic potential flow equations with small perturbation.

$$\Delta p(x - \beta r, r) = p_\infty \frac{\gamma M_\infty^2 F(x - \beta r)}{(2\beta r)^{1/2}}, \quad (1)$$

where  $\Delta p = p - p_\infty$  is the over-pressure associated with the shock wave,  $p_\infty$  is the undisturbed ambient pressure,  $x$  is the axial coordinate which is attached to the flight vehicle,  $r$  is the radius of the source cylinder,  $\gamma$  is the ratio of specific heats,  $M_\infty$  is the flight Mach number,  $\beta = \sqrt{M_\infty^2 - 1}$  is Prandtl–Glauert factor, and

$$F(x) = \frac{1}{2\pi} \int_0^x \frac{A''(\xi)}{(x - \xi)^{1/2}} d\xi \quad (2)$$

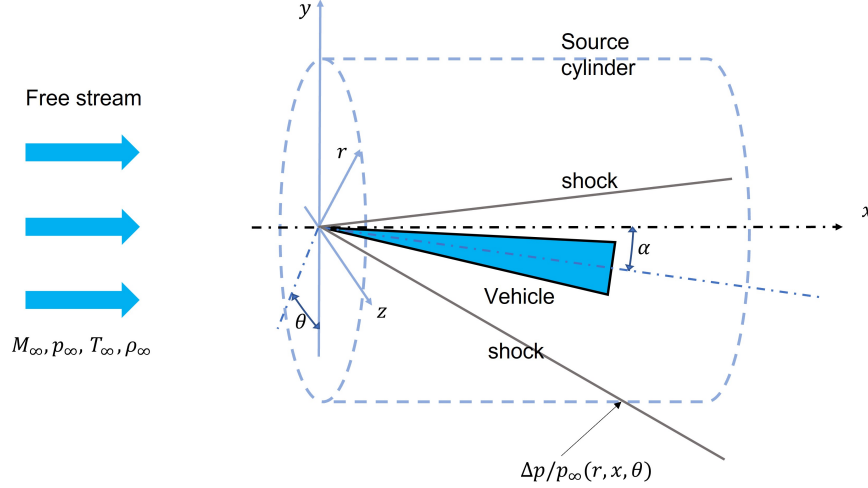
is the F function, where  $A$  is the cross sectional area of the vehicle along cuts aligned with the Mach angle (i.e., slices aligned with iso-surface  $x - \beta r = \text{constant}$ , and hence also called “projected area”), but for slender bodies, it is usually

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**Fig. 1** Coordinate system used in sonic boom problem.

approximated by the cross-sectional area [3].  $A''(\xi)$  denotes a second derivative with respect to  $\xi$ . The F-function theory was extended to lifting geometries later by Walkden [4] by replacing the projected area  $A$  by an effective area

$$A_e(x, \theta) = A_v(x, \theta) + \frac{\beta}{\rho_\infty U_\infty^2} \int_0^x L(\bar{x}, \theta) d\bar{x}, \quad (3)$$

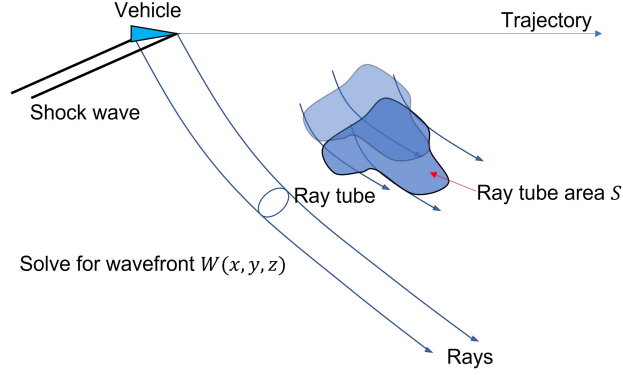
where  $A_v$  is the longitudinal area distribution which is also approximated by cross-sectional area for slender bodies [3], and

$$L(x, \theta) = \cos \theta \int_0^x l(\bar{x}) d\bar{x} \quad (4)$$

is the cumulative (from 0 to  $x$ ) lift projection component in  $-\hat{e}_\theta$  direction, where  $l(x)$  is the lift distribution per unit length. Hayes et al. [5] developed a propagation code by combining F function source and an analytical formulation of ray-tracing (i.e., propagate the signature along the sonic ray tube  $\xi = x - \beta r = \text{constant}$ : see Fig. 2) derived from Fermat's principle. Thomas [5] developed another code using a numerical ray-tracing methodology and propagate parameterized over-pressure signature rather than source expressed by F function. These two codes are the origins of today's ray-trace sonic boom programs [6]. To this day, the most popular way for sonic boom prediction is to determine the near-field over-pressure signature from either computational fluid dynamics (CFD, usually Reynolds-averaged Navier-Stokes, RANS, simulation) or wind-tunnel testing, and propagate the near-field signature using the augmented Burgers equation [7] to determine the mid-field and far-field signatures. Software such as PCBOOM3 [8] from Wyle Lab and US Air Force and sBOOM [9] from NASA Langley research center have hence been developed to address the problem of far-field propagation prediction.

The most recent corresponding practice for the CFD + propagation code methodology are included in the AIAA Sonic Boom prediction workshops [10–12], and the challenges of this method include the high computational cost of CFD [13] and the measurement techniques to avoid interfering the shock structure in supersonic wind tunnels [14]. Thanks to the enhanced computational power enabled by advances in high-performance computing in the recent decade, it is now possible to use full-scale simulations to solve steady or unsteady Euler equations for simple geometries on a relatively coarse mesh (compared to the mesh of near-field full Navier-Stokes simulation) to directly cover the entire range from the near field to the far field [15–17] and good for fundamental research like carpet end diffraction [18]. However, going across more than 5-6 orders of magnitude from  $O(10)$  to  $O(1 \text{ km})$ , the multi-scale nature of the problem remains a big challenge. Leveraging techniques such as Adaptive Mesh Refinement(AMR) will still prove computationally intractable due to the fact that more than 5 layers of AMR would be required to cover the entire range of scales. Limited by resolution of the mesh, it is hence difficult to resolve refined vehicle structures like fins, rendering this approach infeasible for practical low-boom design.

Extending this approach to the hypersonic regime, the process of determining the near-field over-pressure signature  $\Delta p$  (which can be obtained either from CFD or shock-tunnel testing) and propagating it using the augmented Burgers equation still qualifies as a feasible and accurate method. In addition to the intrinsic challenges in supersonic regime,



**Fig. 2 Ray-tracing process in the prediction of sonic boom propagation.**

extremely stretched wall cells induces very small time step size in CFD [19] and rather higher cost in tunnel tests makes the problem even harder. Therefore, a fast and precise theoretical prediction method, if available, will be instrumental to tackle this challenge. However, due to the strong non-linearity in both the flow-structure interaction and extremely strong shocks generated by high freestream Mach number  $M_\infty$ , the normalized maximum over-pressure  $\Delta p_{\max}/p_\infty \sim O(1) - O(10)$ , and hence the assumption of linearity in the Whitham F-function method no longer holds leading to the the method failing at  $M_\infty = 3$  or higher [20]. By revisiting the point explosion problem in 2D under first-order hypersonic limit ( $M_\infty \Delta \rightarrow \infty$  limit for oblique shock relations, where  $\Delta$  is the flow turning angle induced by the structure):

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v)}{\partial r} + \frac{\rho v}{r} = 0, \quad (5)$$

$$\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial r} + \frac{1}{\rho} \frac{\partial p}{\partial r} = 0 \quad (6)$$

$$\frac{\partial (p/\rho^\gamma)}{\partial t} + v \frac{\partial (p/\rho^\gamma)}{\partial r} = 0. \quad (7)$$

with boundary conditions:

$$v|_{r=0} = 0, \quad (8)$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)}{(\gamma - 1) + 2/M^2}, \frac{p_2}{p_1} = 1 + \frac{2\gamma(M^2 - 1)}{(\gamma + 1)}, \frac{v_2}{a_1} = \frac{2}{(\gamma + 1)} [M - (1/M)] \quad (9)$$

$$v = 0, p = p_1, \rho = \rho_1 \text{ for } r > R \quad (10)$$

and energy relation

$$E = 2\pi \int_{r=0}^{r=R(t)} \rho \left[ (e - e_1) + \frac{v^2}{2} \right] r dr, \quad (11)$$

which has a far-field self-similar solution. Tiegerman, advised by Seebass, parameterized the energy impulse  $E$  by the overall flight drag  $D$  and derived a far-field asymptotic solution of maximum over-pressure  $\Delta p_{\max}$  on the ground for a drag-dominant hypersonic vehicle (modeled by a nosed infinite length cone) in homogeneous and isothermal atmosphere [21]:

$$\frac{\Delta p_{\max}}{p_\infty} \Big|_{\text{ground}} = 0.5452 \left( \frac{D^{3/8} e^{-h/8H}}{p_\infty^{3/8} H^{1/4} h^{1/2}} \right), \quad (12)$$

where  $H_0 = 25,000$  ft and  $H$  is the cruise altitude. This theory showed a high accuracy of the maximum over-pressure  $\Delta p_{\max}$  prediction ( $< 5\%$  for both Apollo 15 and 16 re-entry flights) By bridging the near-field strong shock asymptotic solution ( $\Delta p_{\max} \sim r^{-2}$ ) and far-field weak shock asymptotic solution ( $\Delta p_{\max} \sim r^{-3/4}$ ) in 2-D point explosion problem [3], the self-similarity solution of 2D point explosion can be approximated by

$$\frac{\Delta p_{\max}}{p_\infty}(r) = \frac{2\gamma}{(\gamma + 1)} \frac{1}{\beta [(1 + ar^2/R_0^2)^{3/8} - 1]} \quad (13)$$

where  $\alpha = 2.77553$  and  $\beta = 0.8426$  are model constants and  $R_0 = \sqrt{\frac{D}{2\pi p_\infty}}$  is a characteristic length scale of the overall drag  $D$ . This family tree, including Seebass's work [22] and Plooster's work [23], is the so-called blast wave based theory for drag-dominant hypersonic vehicles, which evaluate the shock intensity by analogy [24, 25] to self-similar axisymmetric 2-D point impulse explosion problem and parameterize the energy  $E$  impulse by the overall drag  $D$ . Such a point-based methodology only considers the overall drag but neglects the effects of the specific shape of the vehicle, and lift, which could contribute to a 40% difference in  $\Delta p_{\max}/p_\infty$  in predicting the signature for lifting bodies [26]. To account for geometries and flight conditions which can produce significant lift (e.g., by varying the angle of attack), the predictive theory must account for the vehicle volume and area distribution along the streamwise direction, which requires us to span the point source to a finite length line-distributed source.

To address the limitations of the point-source blast wave theory, in this paper, we developed a line-distributed blast wave (i.e., energy impulse) theory for rapid hypersonic sonic boom prediction, which considers the streamwise distributions ( $d(x)$  and  $l(x)$ ) of both drag and lift. The proposed line-distributed blast wave theory utilizes the Newtonian aerodynamics to estimate both  $d(x)$  and  $l(x)$  and hence gets rid of the expensive CFD simulations or shock tunnel tests for lift and drag information in the state-of-the-art hypersonic sonic boom prediction. Starting from the consistency to the original point-source theory in drag-only cases, various physics including the effects of stratified atmosphere and azimuthal angle are modeled in the extended version of the theory. Test vehicle geometries in this study include a sphere, a nosed cone, a nosed elliptic cone (the first sector of HIFiRE-5 [27]), and a HTV-2 type [28] boost-glide vehicle (BGV) [29] with a sting mount for planned hypersonic shock tunnel test. With respect to the well-acknowledged CFD + propagation method, rapid and accurate prediction is achieved by the newly proposed line-source theory for all the aforementioned geometries (at various angles of attack), which shows the robustness and generality of the new theory.

The remainder of the paper is structured as follows. In Sec. II, the modeling methodology is shown in detail. In Sec. III, near field and far field are discussed separately, both including the validation of the new theory and  $\theta$  dependency of  $\Delta p_{\max}/p_\infty$  within the corresponding  $r$  range. Conclusions are presented in Sec. IV.

## II. Methodology: line-distributed energy impulse formulations

In this paper, we developed a novel line-distributed energy impulse theory for rapid hypersonic sonic boom prediction, which considers the streamwise distributions ( $d(x)$  and  $l(x)$ ) of both drag and lift as an extension of existing point-source blast wave theory:

$$\frac{\Delta p_{\max}}{p_\infty}(r, \theta) = \int_0^{L_0} \phi(x, r; d(x)) \psi(x, r, \theta; l(x)) dx, \quad (14)$$

where  $L_0$  is the length of the vehicle in the streamwise direction,  $\phi(x, r; d(x))$  is the distribution function of drag effect on over-pressure, and  $\psi(x, r, \theta; l(x))$  is the distribution function of lift effect on over-pressure. This new formulation is an integration over a finite length line-distributed source rather than a point source. Due to the presence of an additional streamwise dimension in the formulation, it naturally considers the origin of the lift (i.e., area and volume distribution in the streamwise direction) and hence is suitable for lifting geometries.

### A. Formulation of drag effect distribution function $\phi(x, r; d(x))$

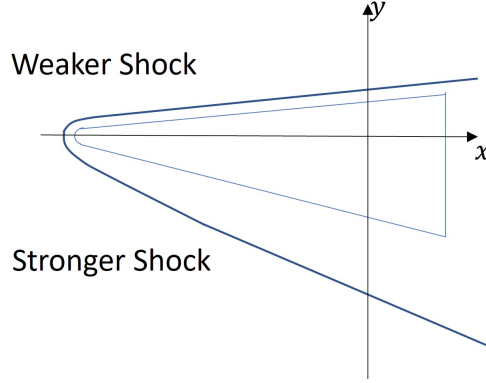
For drag-only scenarios, the lift distribution function becomes  $l(x) = 0$  and hence the lift effect distribution function is reduced to  $\psi(x, r; l(x) = 0) = 1$ . Then the proposed line-distributed formulation reduces to:

$$\frac{\Delta p_{\max}}{p_\infty}(r) = \int_0^{L_0} \phi(x, r; d(x)) dx. \quad (15)$$

To achieve consistency with the point-source theory [23] (i.e., Eq. 13) for drag-only scenarios, we must have:

$$\int_0^{L_0} \phi(x, r; d(x)) dx = \frac{\Delta p_{\max}}{p_\infty}(r) = \frac{2\gamma}{(\gamma + 1)} \frac{1}{\beta \left[ (1 + \alpha r^2/R_0^2)^{3/8} - 1 \right]}, \quad (16)$$

where  $R_0 = \sqrt{\frac{D}{2\pi p_\infty}}$  is a characteristic length scale of the overall drag  $D$ . This equation is the foundation to derive the formulation of drag effect distribution function  $\phi(x, r; d(x))$ . To do so, it is critical to convert the dependence of maximum over-pressure  $\Delta p_{\max}$  on overall drag  $D$  to the dependence on drag distribution  $d(x)$  along the streamwise direction  $x$ . For convenience, we define the cumulative drag in interval  $[0, x]$  as  $D'(x) = \int_0^x d(\bar{x}) d\bar{x}$ , and hence the



**Fig. 3 Intensity of the shock induced the strong lifting effect of hypersonic vehicles**

overall drag  $D = D'(L_0)$ , where  $L_0$  is the streamwise length of the vehicle geometry. Now consider  $L_0$  as a variable rather than a constant, and take derivative of Eq. 13 with respect to  $L_0$ , then we can derive a formulation of  $\phi(x, r; d(x))$ :

$$\phi(x, r; d(x)) = \frac{3\alpha r^2 (2\pi p_\infty) d(x)}{8\beta \left[ \left( 1 + \frac{\alpha r^2 (2\pi p_\infty)}{D'(x)} \right)^{\frac{3}{8\beta}} - 1 \right]^2 D'^2(x) \left( 1 + \frac{\alpha r^2 (2\pi p_\infty)}{D'(x)} \right)^{\frac{5}{8\beta}}} \quad (17)$$

To compute the emerged new functions  $d(x)$  and  $D'(x)$  without the expensive CFD simulations, Newtonian aerodynamics is used:

$$d(x) = \alpha_v \left( \frac{1}{2} \rho_\infty U_\infty^2 \right) \int_0^{2\pi} c_p(x, \theta) \max[\cos\delta(x, \theta), 0] dA(x, \theta), \quad (18)$$

where  $\alpha_v$  is the viscous correction factor (typical value range: 0.9 ~ 1.1), pressure coefficient  $c_p(x, \theta) = 2\cos^2\delta$ ,  $\cos\delta(x, \theta) = -\mathbf{i} \cdot \mathbf{n}(x, \theta)$ ,  $\mathbf{n}(x, \theta)$  is the local unit outward normal vector of the body surface,  $\mathbf{i}$  is the unit vector of axial  $x$ -direction (i.e., streamwise direction), and  $A(x, \theta)$  is the body surface area distribution function. The integration for  $d(x)$  is computed numerically. Then the cumulative drag  $D'(x) = \int_0^x d(\bar{x}) d\bar{x}$  is integrated numerically.

### B. Formulation of lift effect distribution function $\psi(x, r, \theta; l(x))$

For hypersonic lifting vehicles, the primary lifting force comes from the pressure difference between the top and bottom surfaces, which originates from the special design that can exhibit the difference in the shock intensity on the top and bottom of the vehicle as shown in Fig. 3 and confirmed by Fig. 21. To account for such lift effects, an amplifier function  $\psi(x, r, \theta; l(x))$  should be applied to the drag effect distribution function  $\phi(x, r; d(x))$  as in Eq. 14.

For scenarios with weak lift (i.e., lift-to-drag ratio  $L/D \ll 1$ ), based on the Taylor expansion with respect to  $L/D$ , this amplifier  $\psi(x, r, \theta; l(x))$  can be linearly approximated as:

$$\psi(x, r, \theta; l(x)) \approx 1 + (L/D)\psi'(x, r; l(x)), \quad (19)$$

where the second term (i.e., the linear term) on the right-hand-side (RHS) represents the lift effect, and  $L$  and  $D$  are the overall lift and drag, respectively, estimated by Newtonian aerodynamics. Note that this formulation could be made valid even for scenarios with strong lift (i.e.,  $L/D \geq 1$ ): a constant  $C_{L/D}$  can be chosen such that  $C_{L/D}L/D \ll 1$ , and the Taylor expansion of  $\psi(x, r, \theta; l(x))$  with respect to  $C_{L/D}L/D$  can be applied to get  $\psi'(x, r; l(x))$ , that absorbs the constant  $C_{L/D}$ . Hence the linearized formulation becomes the same as Eq. 19 but is valid even if  $L/D \geq 1$ .

$\psi'(x, r)$  is a shape coefficient which represents the transfer of lift effect to maximum over-pressure. By mimicking the formulation of drag density distribution function  $d(x)$ ,  $\psi'(x, r; l(x))$  can be modeled as:

$$\psi'(x, r; l(x)) = \int_0^{2\pi} \max[\cos\eta(x, \theta), 0] \psi'_*(x, r; l(x)) dA(x, \theta) = \int_0^{2\pi} \max[-\mathbf{j} \cdot \mathbf{n}(x, \theta), 0] \psi'_*(x, r; l(x)) r d\theta, \quad (20)$$

where  $\mathbf{j}$  is the unit vector of vertical  $y$ -direction and  $\psi'_*(x, r; l(x))$  is an unclear scaling factor function to account for the potential complexity (e.g., the possible inaccuracy of Newtonian aerodynamics in lift prediction). Without more

information, for simplicity,  $\psi'_*(x, r; l(x))$  is designed to make  $\psi'(x, r; l(x))$  dimensionless:

$$\psi'_*(x, r; l(x)) = \frac{1}{\int_0^{2\pi} r d\theta}. \quad (21)$$

### C. Drag corrections: effect of finite length clipping and viscous drag

Based on the Newtonian aerodynamic assumption, the contribution of the ambient pressure  $p_\infty$  to the surface pressure coefficient  $c_p(x, \theta)$  cancels, due to the fact that it is counted in both the upwind side and downwind side. Such assumption is valid for any flight scenarios without significant flow separation, typically vehicle geometries with a gradually changing curvature: e.g., a slender body and a sphere. Such an assumption is invalid for any clipped body (e.g., a finite-length cone, the first section of HiFiRE-5, and the boost glide vehicle (BGV) geometry used later in this paper), due to the presence of strong flow separation and the presence of a very low pressure bubble (i.e., re-circulation zone) in the wake of the geometry (e.g., see Figs. 11 and 21). Using the fact that the pressure in the separation bubble is much lower than  $p_\infty$  (often  $< 10\% p_\infty$ ), it is reasonable to count the ambient pressure contribution in the upwind side and neglect the ambient pressure contribution in the downwind side for clipped vehicle geometries:

$$c_p = c_{p, \text{Newtonian}} + c_{p_\infty} \delta_{\text{clip}} = 2 \cos^2 \delta + \frac{2p_\infty}{\rho_\infty U_\infty^2} \delta_{\text{clip}}, \quad (22)$$

where  $\delta_{\text{clip}} = 1$  if  $\min(\mathbf{n} \cdot \mathbf{n}_{\text{neighbor}} \leq 0)$  (i.e., a clipped body) and  $\delta_{\text{clip}} = 0$  otherwise.

Apart from this, CFD simulations indicate that among all the tested geometries, the viscous drag is about 10% of the form drag, an additional viscous drag correction  $\alpha_v = 1.1$  is recommended here.

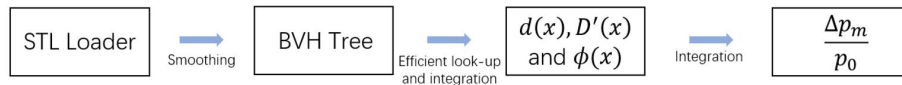
According to Table. 1, the aforementioned drag corrections exhibits the significant drag prediction improvement for all the clipped vehicle geometries considered in this study. These corrections can impact both drag effect distribution function  $\phi(x, r; d(x))$  and lift effect distribution function  $\psi(x, r, \theta; l(x))$ , especially the latter one due to the linear dependence on  $L/D$ .

**Table 1 Drag prediction given by CFD, the original Newtonian aerodynamics, and the new formulation with drag corrections for several clipped vehicle geometries.**

| Geometry | $D_{\text{old}}$ (N) | $D_{\text{new}}$ (N) | $D_{\text{CFD}}$ (N) |
|----------|----------------------|----------------------|----------------------|
| Cone     | 7.11                 | 26.40                | 29.40                |
| HiFiRE-5 | 408.65               | 890.29               | 870.00               |
| BGV      | 62.24                | 125.42               | 136.37               |

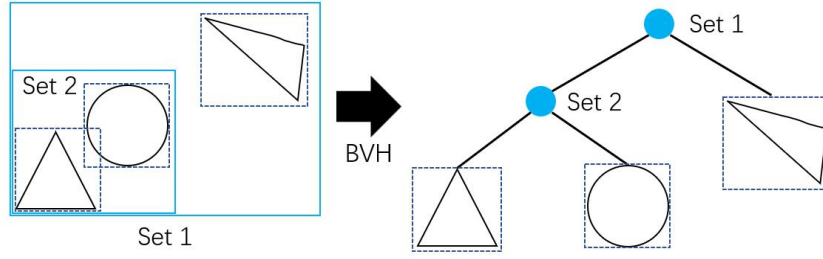
### D. Acceleration of facet searching

The flow chart of the newly proposed algorithm is summarized in Fig. 4. To make the integration over the geometry surface (i.e. determination of  $\phi(x)$  and  $\psi(x)$ ) simpler and more flexible to complex geometries, the STL format (triangulated surface) is selected as the input format. As a result, for each interval  $(x, x + \Delta x)$ , a search algorithm among the whole triangle set is needed to find all triangles overlapping with the interval for the local evaluation of  $d(x)$ ,  $\phi(x)$  and  $\psi(x)$ . In practice, a high-quality STL file generated by CAD software usually including  $O(10^3) - O(10^5)$  facets is



**Fig. 4 A simple flowchart of the algorithm process**

used. The corresponding  $O(N)$  searching is then quite time-costly ( $O(10^4)$  times searching during the whole process is needed for accurate output). Due to the similarity of the process to ray-tracking in rendering, bounding volume hierarchy (BVH), a tree-structure illustrated in Fig. 5, which is commonly used to accelerate the ray-tracking process is employed to accelerate the searching process from  $O(N)$  to  $O(\log(N))$ .



**Fig. 5** brief illustration of BVH tree

### III. Results and Discussion

To validate proposed formulations, the standard combination of CFD and wave propagation was employed to generate the benchmark data. The hypersonic CFD simulations were conducted using US3D code [30–32], which was designed for the simulation of aerothermodynamics and highly compressible hypersonic flows involve complex multi-disciplinary physics and has been well-validated and widely used in the hypersonic community. The wave propagation simulations were conducted using the NASA sBOOM software [9], which has been well-validated and widely used in the sonic boom community.

#### A. Test vehicle geometries

Test vehicle geometries in this study include a sphere (Sec. III.A.1), a nosed cone (Sec. III.A.2), a nosed elliptic cone (the first sector of HIFiRE-5 [27]: Sec. III.A.3), and a HTV-2 type [28] boost-glide vehicle (BGV: Sec. III.A.4) [29] with a sting mount for planned hypersonic shock tunnel tests in CUBRC, Inc.

##### 1. Sphere

As a fundamental study, it is beneficial to start with a simple geometry like sphere. Using US3D CFD code, we simulated a hypersonic sphere with diameter  $D_0 = 1$  m at  $M_\infty = 5$  and  $M_\infty = 10$  at a cruise altitude  $H = 20$  km condition (5417.7 Pa and 216.65 K) on a mesh with  $\sim 2.4$  million cells for a quarter of the domain. Such mesh is already fine enough for the near-field over-pressure data collection based on the state-of-the-art in the literature [33].

##### 2. Nosed cone

Second, we simulated a nosed cone geometry, with a semi-vertex angle of 6 degree, a length of  $L_0 = 0.3$  m, and a nose with 2.5 mm radius. The free stream condition for the nosed cone cases are  $M_\infty = 5, 6$  and  $10$ ,  $\rho_\infty = 0.0113 \text{ kg/m}^3$ , and  $T_\infty = 300$  K, which is an approximated shock tunnel condition. For the quarter mesh (i.e. for  $\alpha = 0$ ), mesh size is  $\sim 5$  million cells. For  $\alpha \neq 0$  cases, mesh size is  $\sim 15$  million cells.

##### 3. Nosed elliptic cone (first sector of HIFiRE-5)

To evaluate the azimuthal angle  $\theta$  effect, we also simulated a nosed elliptic cone (the first sector of HIFiRE-5 [27]). The half-angle of the elliptic cone test article in the minor axis (x-y) plane is seven degrees, and 13.797 degrees in the major axis. The nose tip cross-section in the minor axis is a 2.5 mm radius circular arc, tangent to the cone ray describing the minor axis, and retains a 2:1 elliptical cross-section to the stagnation point. The elliptic cone major axis diameter is 431.8 mm at the base, and its length  $L_0 = 861$  mm. For  $\alpha = 0$  cases, mesh size is  $\sim 6.5$  million cells, and for  $\alpha \neq 0$  cases, mesh size is  $\sim 13$  million cells. The flight condition is  $M_\infty = 5$  and cruise altitude  $H = 20$  km.

##### 4. Boost-glide vehicle (BGV) with a sting mount

Finally, we simulated a HTV-2 type [28] boost-glide vehicle (BGV) [29] with a sting mount for planned hypersonic shock tunnel tests in CUBRC, Inc. The geometry length is about  $L_0 = 0.3657$  m or 1.2 ft and the length of the sting is about 0.2 m. Zero angle of attack is used in the simulation. The flight condition is  $M_\infty = 5$  and  $M_\infty = 10$  at cruise altitude  $H = 20$  km and  $H = 30$  km, under such conditions,  $L/D = 2.48$ . The mesh for this geometry is  $\sim 8$  million

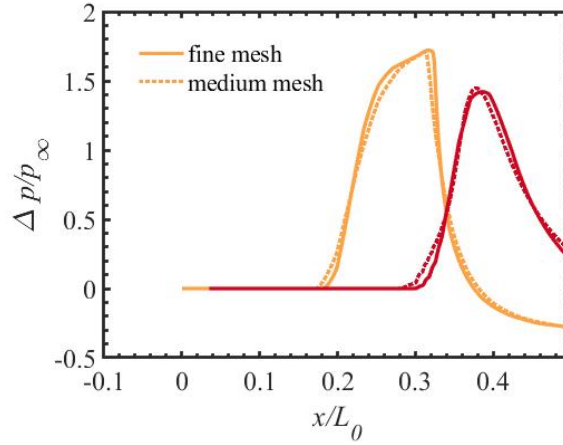


because of the biased interest(side with expansion effect is almost excluded from the domain).

## B. Near-field over-pressure signatures: theories vs. CFD

### 1. Sensitivity of CFD prediction to grid resolution and cruise altitude

Due to the high speed, Reynolds number of hypersonic flow is usually high and the viscous effect is almost limited within the boundary layer, which means that for the off-body over-pressure  $\Delta p$  extraction, a relative coarse mesh is sufficient. In all of the CFD simulation cases in this study, mesh size is around 10 million grid cells with a refined boundary layer mesh. To test the sensitivity of over-pressure  $\Delta p$  to grid resolution, a cone with zero angle of attack at a hypersonic inflow condition ( $M_\infty = 6$ ,  $\rho_\infty = 0.0113 \text{ kg/m}^3$ ,  $T_\infty = 300 \text{ K}$ ) is used for grid convergence test. The sampled over-pressure  $\Delta p$  data at various  $r$  is given in Fig. 6, which shows a negligible difference between the medium mesh ( $\sim 10$  million grid cells) and fine mesh ( $\sim 15$  million grid cells) which has a factor of two refinement in the bulk region where the over-pressure data is sampled.



**Fig. 6** Quantitative comparison of normalized over-pressure  $\Delta p/p_\infty$  at  $r/L = 0.13$  (orange) and  $r/L = 0.2$  (dark red) between two different meshes for a cone with zero angle of attack at a hypersonic inflow condition ( $M_\infty = 6$ ,  $\rho_\infty = 0.0113 \text{ kg/m}^3$ ,  $T_\infty = 300 \text{ K}$ ).

Regarding the sensitivity to cruise altitude, two sets of simulation were conducted for the BGV configuration (with a sting mount) at zero angle of attack and two cruise altitudes  $H = 20 \text{ km}$  and  $H = 30 \text{ km}$ . The results at different  $r/L_0$  with two free stream Mach numbers  $M_\infty = 5$  and  $M_\infty = 10$  are given in Fig. 7. Assuming ideal gas, the results indicate that normalized over-pressure  $\Delta p/p_\infty$  is not sensitive to the cruise altitude  $H$  once Mach number is fixed.

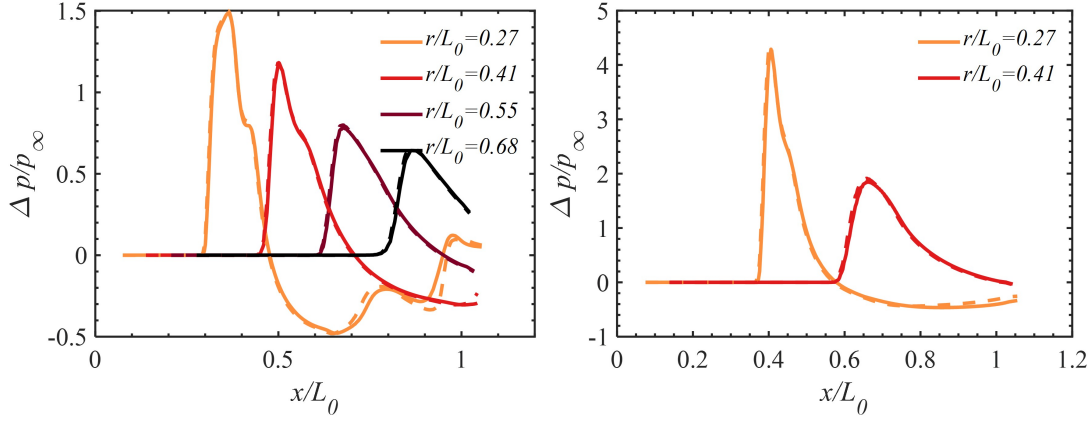
### 2. Scenarios without lift: validation of drag effect distribution function $\phi(x, r; d(x))$

Hypersonic sphere is the first case without lift, and its corresponding over-pressure  $\Delta p$  contour plots at  $M_\infty = 5$  and  $M_\infty = 10$  are given in Fig. 8. As can be seen from the contours, doubling the Mach numbers provide five times stronger over-pressure  $\Delta p$ , but its spatial distribution is insensitive to Mach number.

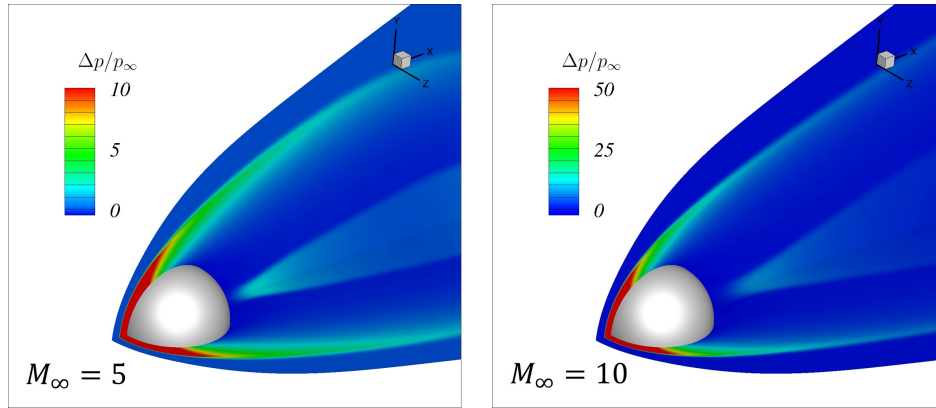
From the CFD simulation data, we can extract the over-pressure signature  $\Delta p$  at various locations of  $r/D_0$  where  $D_0$  is the diameter of the sphere (i.e., the vehicle length  $L_0$  of the sphere), as shown in Fig. 9. Due to the axis-symmetric nature and for convenience, the data is extracted at  $z = 0$ . The drop-down of over-pressure  $\Delta p$  at small  $r/D_0$  comes from the recirculation zone (i.e., the wake) behind the sphere. Further away from the body, both the peak value and drop-down decay very quickly (probably exponentially).

To validate the drag effect distribution function  $\phi(x, r; d(x))$ , we applied the line-distributed energy impulse formulation to the sphere cases, as shown in Fig. 10. The comparison target here is the normalized maximum over-pressure  $\Delta p_{\max}/p_\infty$  extracted from the CFD data shown in Fig. 9. In addition, based on the consistency constrain of the derivation (i.e., Eq. 16), our new line-distributed formulation should agree with the classical point-source blast wave theory when there is no lift. For this reason, we also include the prediction result of the classical point-source blast wave theory in Fig. 10, which directly use the overall drag  $D$  extracted from CFD. As shown in Fig. 10, predictions





**Fig. 7** Quantitative comparison of normalized over-pressure  $\Delta p/p_\infty$  between cruise altitude  $H = 20$  km (solid line) and  $H = 30$  km (dashed line) for the BGV configuration (with a sting mount) at zero angle of attack and various locations and Mach numbers:  $M_\infty = 5$  (left) and  $M_\infty = 10$  (right).



**Fig. 8** Normalized over-pressure  $\Delta p/p_\infty$  field of a sphere at two hypersonic Mach numbers:  $M_\infty = 5$  (left) and  $M_\infty = 10$  (right).

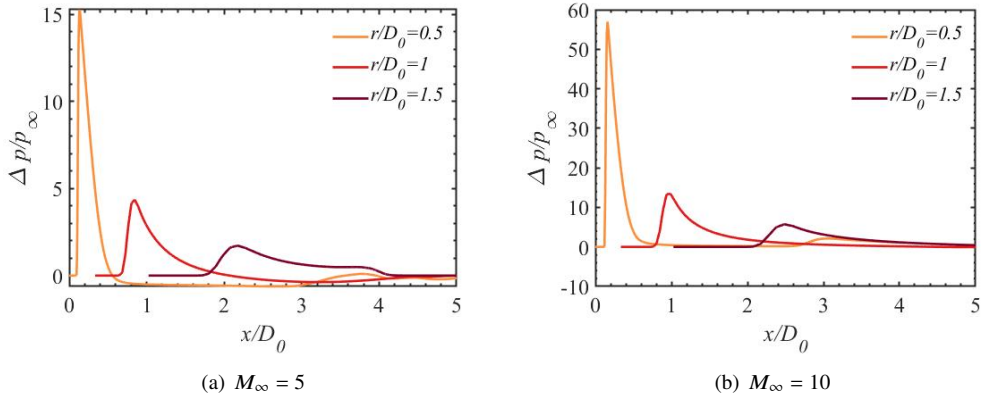
from the line-distributed formulation match the classic point-source blast wave theory results, as expected, and their difference comes from the error of Newtonian aerodynamics. This indicates that using Newtonian aerodynamics for force approximation in the line-source formulation is valid, at least for spheres. Also, both of formulations match well with the CFD data, and hence the validation is successful.

Nosed cone with zero angle of attack (i.e.,  $\alpha = 0$ ) is another case without lift, the corresponding flow field is shown in Fig. 11. A clear low-pressure zone appears right behind the base of the cone, because of the strong flow separation zone in the wake of the clipped geometry. This observation is the original motivation of the drag correction based on the effect of finite length clipping in Sec. II.C. As shown later in Fig. 15, the drag correction significantly improves the drag  $D$  prediction and hence also the lift-to-drag ratio  $L/D$  prediction.

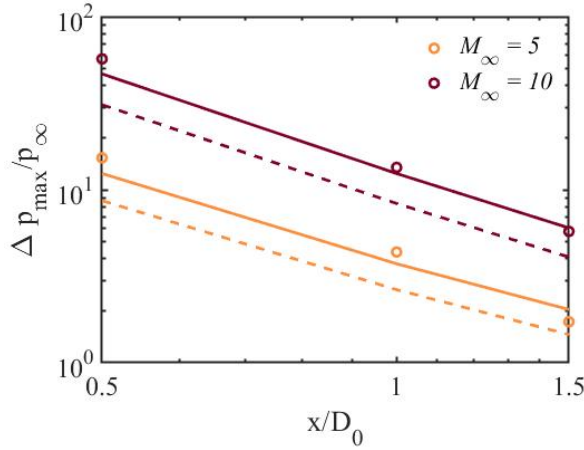
The over-pressure  $\Delta p$  signatures are extracted from the CFD data and shown in Fig. 12. With a much stronger wake flow separation compared to the sphere (at the same hypersonic Mach numbers), the nosed cone signatures contain a much larger drop-down of over-pressure  $\Delta p$ , and such a drop-down with negative values does not disappear even at relatively large  $r/D_0$  (e.g.,  $r/D_0 = 1.333$ ).

To further validate the drag effect distribution function  $\phi(x, r; d(x))$ , we applied both the line-distributed energy impulse formulation (i.e., ‘new’) and classical point-source blast wave theory (i.e., ‘old’) to the nosed cone cases, as shown in Fig. 13. The comparison target here is the normalized maximum over-pressure  $\Delta p_{max}/p_\infty$  extracted from the CFD data shown in Fig. 12.

Predictions from both the new and old theory are close to the CFD data, which again validates our new proposed



**Fig. 9** Normalized over-pressure  $\Delta p/p_\infty$  vs. normalized streamwise distance  $x/D_0$  of a sphere at two hypersonic Mach numbers: (a)  $M_\infty = 5$  and (b)  $M_\infty = 10$ .



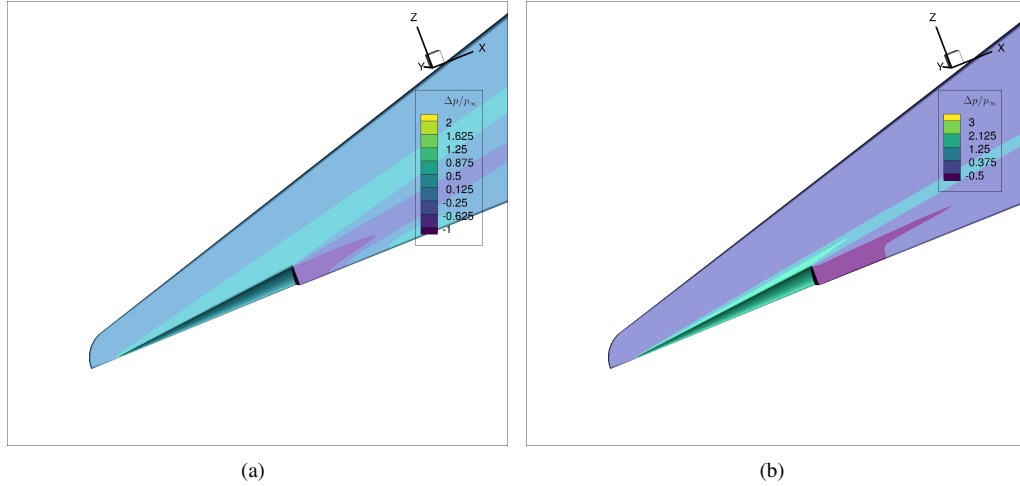
**Fig. 10** Comparison of normalized maximum over-pressure  $\Delta p_{\max}/p_\infty$  between the CFD data (circles), classical point-source blast wave theory (i.e., “Old Formulation”: dashed lines), and new line-distributed energy impulse formulation (i.e., “New Formulation”: solid lines) for a sphere at two hypersonic Mach numbers:  $M_\infty = 5$  (yellow) and  $M_\infty = 10$  (purple).

theory. According to the CFD data, at relatively large  $r/L_0$ , the normalized maximum over-pressure  $\Delta p_{\max}/p_\infty$  at  $M_\infty = 10$  is smaller than at  $M_\infty = 5$ , which is counter-intuitive and could be caused by the coarse CFD mesh near the end of the domain. Both theories do not follow this potentially incorrect trend.

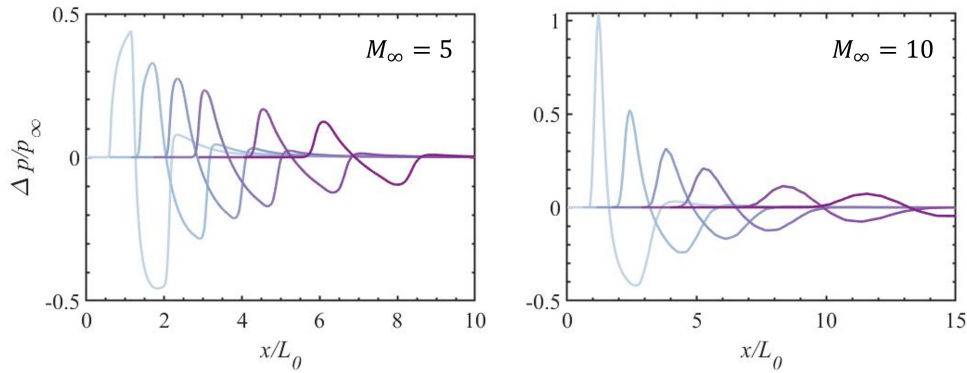
### 3. Scenarios with lift and/or azimuthal angle $\theta$ effect: validation of lift effect distribution function $\psi(x, r, \theta; l(x))$

For the same aforementioned nosed cone, we can incline it (i.e., nonzero angles of attack) to generate lift. Specifically, at the free stream condition mentioned in Sec. III.A.2,  $M_\infty = 6$ , and two angles of attack of  $\alpha = 5^\circ$  and  $\alpha = 10^\circ$ , the over-pressure  $\Delta p$  field is shown in Fig. 14. High pressure regions can be observed below the nosed cone which is the source of lift  $L$ , and the pressure increases with  $\alpha$ . As a result, the lift-to-drag ratio  $L/D$  of the nosed cone increases from 1.58 to 2.08 when  $\alpha$  is increased from  $5^\circ$  to  $10^\circ$ .

The lift-to-drag ratio  $L/D$  of the nosed cone is plotted as a function of angle of attack  $\alpha$  is given in Fig. 15. As shown in the figure,  $L/D$  could vary from 0 at  $\alpha = 0^\circ$  to 2.08 at  $\alpha = 10^\circ$ , which grant us a wide spectrum of data to validate the lift effect distribution function  $\psi(x, r, \theta; l(x))$ . The lift  $L$  prediction from Newtonian aerodynamics is reasonably accurate, but the drag  $D$  prediction from Newtonian aerodynamics contains significant error (for the reason mentioned in Sec. II.B) and hence result in significant error in  $L/D$  prediction. As shown in the figure, the drag  $D$



**Fig. 11** Normalized over-pressure  $\Delta p/p_\infty$  field of a nosed cone at zero angle of attack (i.e.,  $\alpha = 0$ ) and two hypersonic Mach numbers:  $M_\infty = 5$  (left) and  $M_\infty = 10$  (right).



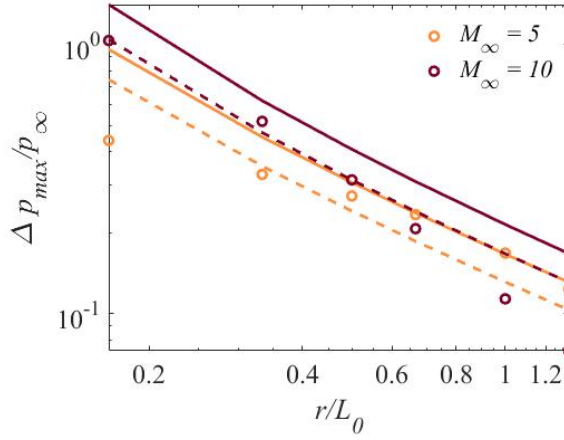
**Fig. 12** Normalized over-pressure  $\Delta p/p_\infty$  field of a nosed cone at  $r/L_0 = 0.167, 0.333, 0.5, 0.667, 1$ , and  $1.333$  (from light color to dark color) and two hypersonic Mach numbers:  $M_\infty = 5$  (left) and  $M_\infty = 10$  (right).

corrections in Sec. II.B significantly improve the both  $D$  prediction and  $L/D$  prediction.

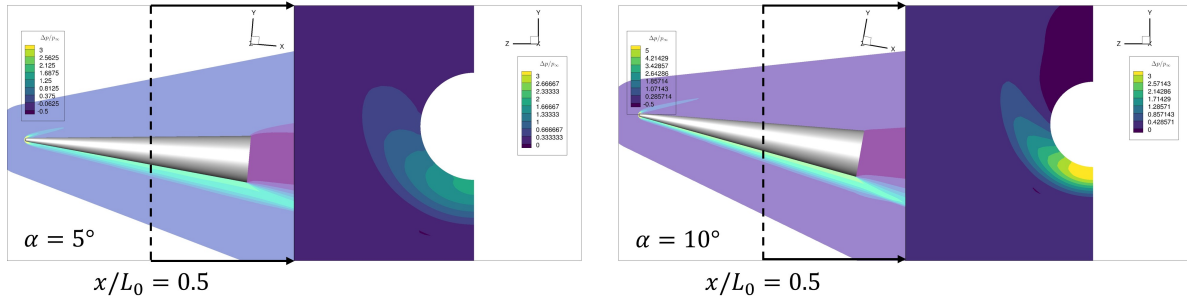
Over-pressure  $\Delta p$  signatures of the nosed cone at different  $\alpha$  is then sampled at three different near-field locations  $r/L_0 = 0.27$ ,  $r/L_0 = 0.4$  and  $r/L_0 = 0.53$ , and the maximum over-pressure  $\Delta p_{max}$  data is extracted, which is then compared with the theory prediction in Fig. 16. At relatively small  $\alpha$  (e.g.,  $0^\circ$ - $1^\circ$ ), prediction from the line-distributed energy impulse theory agrees well with the CFD data and hence the theory and lift effect distribution function  $\psi(x, r, \theta; l(x))$  are well-validated. However, when  $\alpha$  becomes large (e.g.,  $5^\circ$ - $10^\circ$ ), error of the theory increases, because of the significant azimuthal angle  $\theta$  effect (the cross section perpendicular to the free stream flow becomes very elliptic at large  $\alpha$ ) and a larger low-pressure bubble (i.e., recirculation zone) in the wake, as shown in Fig. 14.

It is worthy to check our drag correction for all of the clipping geometries, here we also computed the  $L$ ,  $D$  and  $L/D$  for the first section of HiFiRE-5 [27]. As shown in Fig. 17. Satisfactory prediction for all three geometries is achieved. The reasoning behind the selection of the HiFiRE-5 geometry is that it is the simplest configuration with asymmetry even at zero angle of attack (i.e.,  $\alpha = 0$ ). Hence HiFiRE-5 is ideal for the effect of the combining effect of azimuthal angle  $\theta$  and the radius  $r$  for modeling insights. Two slice of the  $\Delta p/p_\infty$  field is given in Fig. 18. In the upstream ( $x/L_0 = 0.94$ ), a strong  $\theta$  dependence is observed. However, in the downstream ( $x/L_0 = 1.71$ ), the normalized over-pressure  $p/p_\infty$  contour is closer to a set of concentric circles which are independent of  $\theta$ .

To quantitatively analyze the azimuthal angle  $\theta$  effect and the decay of the effect, normalized over-pressure  $\Delta p/p_\infty$  profiles are evaluated at a series of  $\theta$  and presented in Fig. 19. The azimuthal angle  $\theta$  effect decays asymptotically with



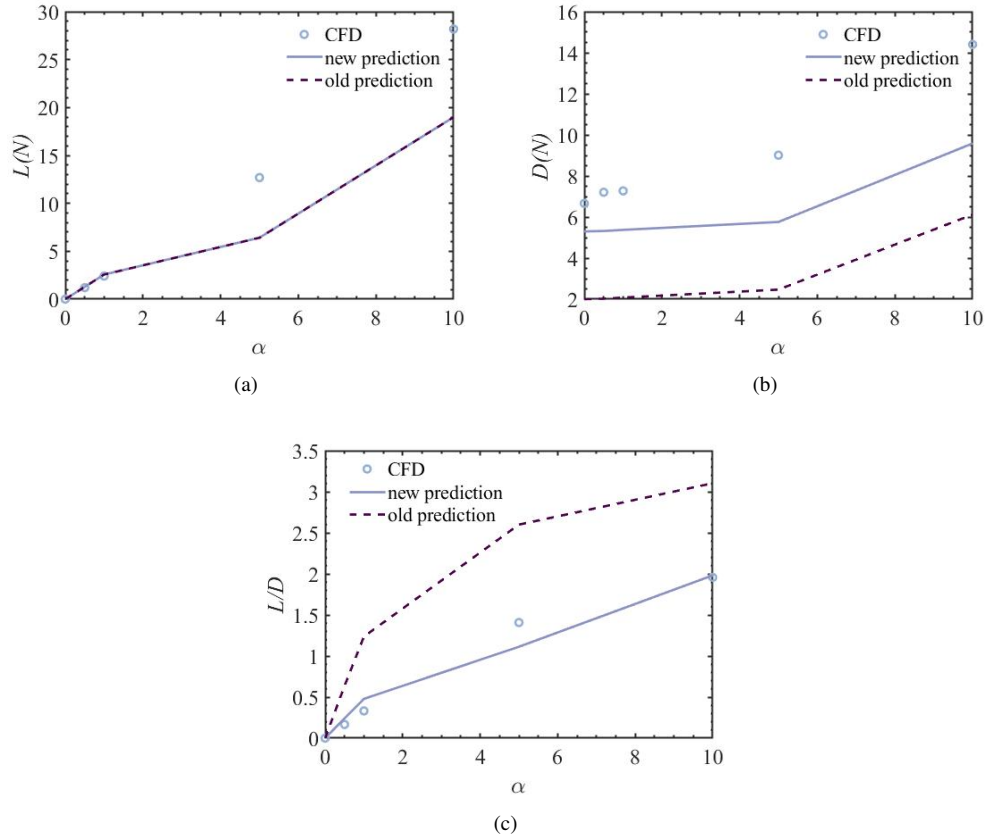
**Fig. 13** Theoretical prediction from new (solid lines) and old (dashed lines) theories for a nosed cone at two hypersonic Mach numbers ( $M_\infty = 5$  and  $M_\infty = 10$ ), compared to CFD data (circles).



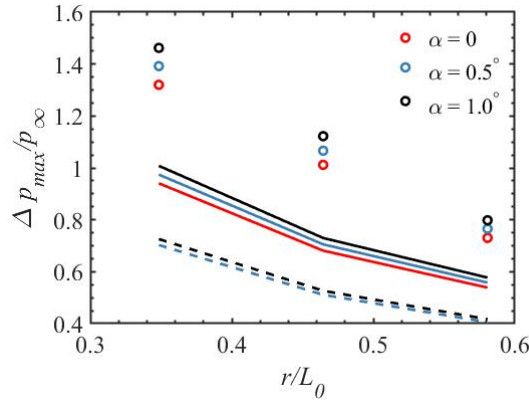
**Fig. 14** Normalized over-pressure  $\Delta p/p_\infty$  field in the slice of the symmetric plane of the nosed cone at  $M_\infty = 6$  with angle of attack  $\alpha = 5^\circ$  (left) and  $\alpha = 10^\circ$  (right).

a characteristic length  $L^* \sim L_0$ . For example, the  $\theta$  effect on maximum over-pressure  $\Delta p_{\max}$  decays from  $\sim 50\%$  at  $r/L_0 = 0.3$  to  $\sim 20\%$  at  $r/L_0 = 0.5$ .

Our proposed line-distributed blast wave theory (using Newtonian aerodynamics) is compared with the classical point-source blast wave theory (using the drag from the CFD) in the near field, as shown in Fig. 20. Although neither of the two theories agree perfectly with the CFD data, our new line-distributed theory with drag corrections provides better predictions than the classical point-source theory, which does not consider the lift effect. This is an impressive performance considering only Newtonian aerodynamics instead of accurate CFD data is used in the line-distributed theory. At zero angle of attack (i.e.,  $\alpha = 0$ ), there is no lift and hence no lift effect. The error of the theories must come from strong near-field effect or strong azimuthal angle  $\theta$  effect on normalized maximum over-pressure  $\Delta p_{\max}/p_\infty$ .

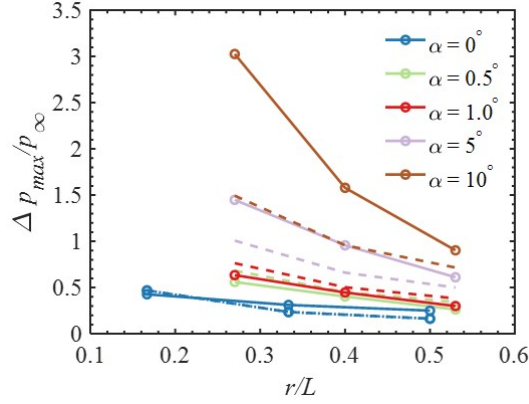


**Fig. 15** (a) Lift  $L$ , (b) drag  $D$ , and (c) lift-to-drag ratio  $L/D$  predicted by CFD, proposed line-distributed theory ('new'), and classical point-source theory (i.e., 'old'), as functions of attack angle  $\alpha$  for the nosed cone at  $M_\infty = 6$ .



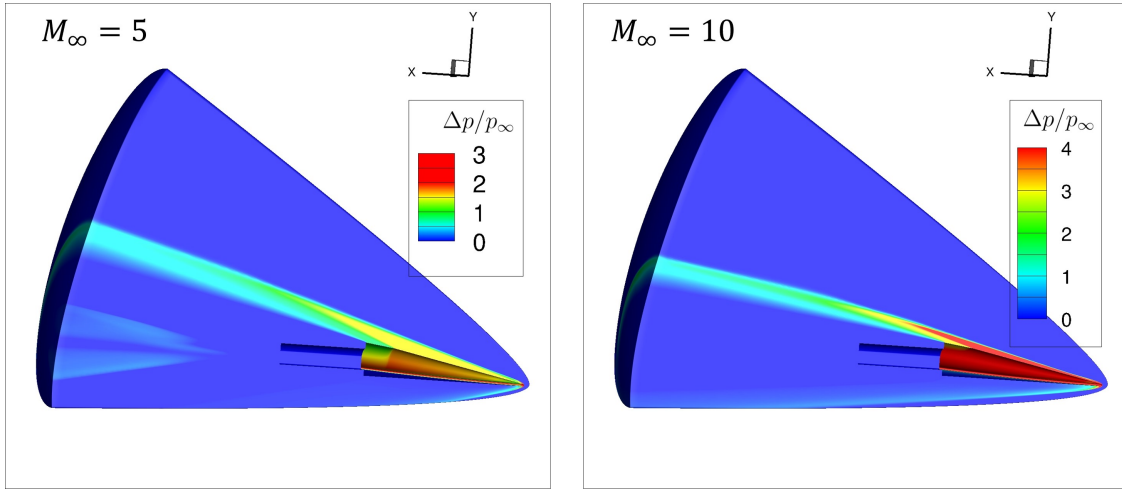
**Fig. 20** Normalized maximum over-pressure  $\Delta p_{\max}/p_\infty$  of a nosed elliptic cone (the first sector of HIFiRE-5) at different angles of attack  $\alpha$ : CFD (circles), proposed line-distributed theory (solid lines), classical point-source theory (dashed lines).

Another configuration with strong lift and azimuthal angle  $\theta$  effect is boost-glide vehicle (BGV). we simulated a HTV-2 type [28] boost-glide vehicle (BGV) [29] at zero angle of attack (i.e.,  $\alpha = 0$ ). The normalized over-pressure  $\Delta p/p_\infty$  fields at both  $M_\infty = 5$  and  $M_\infty = 10$  are shown in Fig. 21. In the figure, the configuration is flipped upside-down to better show the strong high pressure regions below the BGV, which give a large lift-to-drag ratio  $L/D \approx 2.48$  even



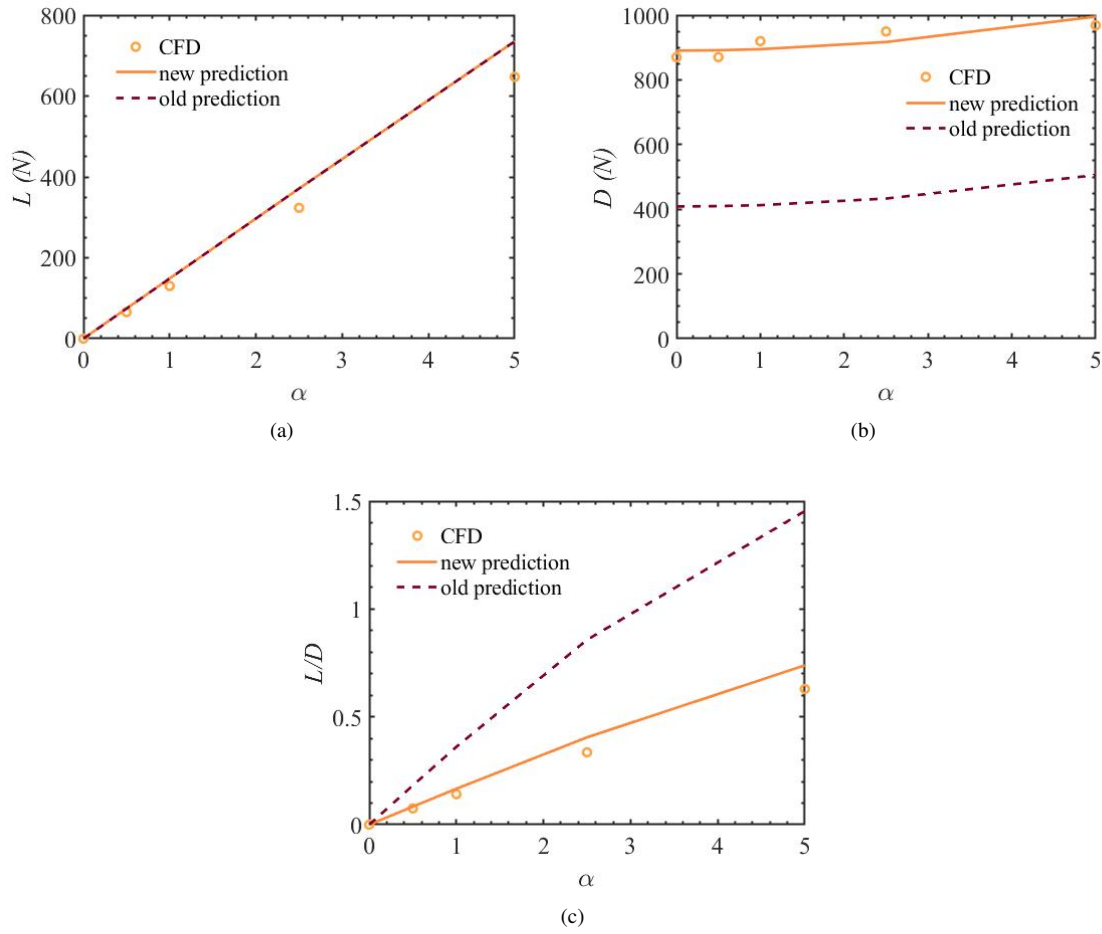
**Fig. 16** Normalized maximum over-pressure  $\Delta p_{\max}/p_{\infty}$  of nosed cone at several angles of attack  $\alpha$  at  $r/L_0 = 0.27$ ,  $r/L_0 = 0.4$  and  $r/L_0 = 0.53$ : predictions from CFD (circle-solid lines) and line-distributed theory (dashed lines).

at  $\alpha = 0$ . Comparing to previous configurations, the over-pressure of BGV is less sensitive to Mach number and the change from  $M_{\infty} = 5$  to  $M_{\infty} = 10$  is only  $\sim 30\%$ .

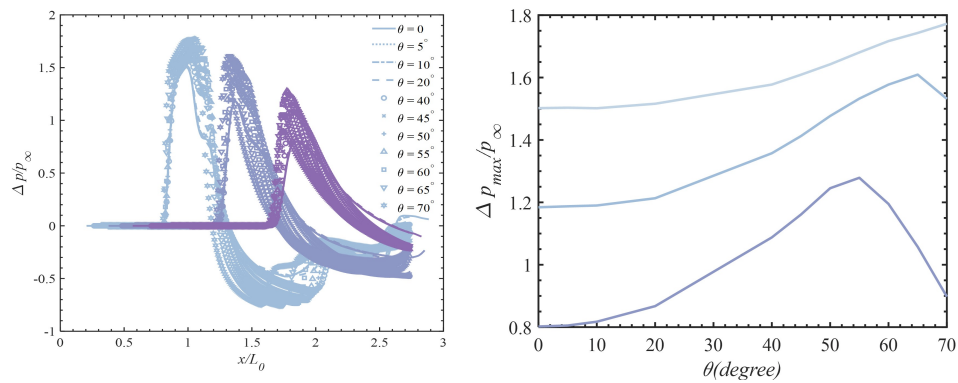


**Fig. 21** Normalized over-pressure  $\Delta p/p_{\infty}$  field in the slice of the symmetric plane of a boost-glide vehicle (BGV) with a sting mount at zero angle of attack:  $M_{\infty} = 5$  (left) and  $M_{\infty} = 10$  (right).

Due to the presence of the strong lift for the BGV geometry, a non-uniform and non-monotonic distribution of  $\Delta p_{\max}/p_{\infty}$  over  $\theta$  was observed in Fig. 22. It is shown that there is a peak generated by the shape close to  $\theta = 90^\circ$  in the near field, and moving inward when  $r$  grows: from  $\theta_{\text{peak}} = 70^\circ$  at  $r/L_0 = 0.27$  to  $\theta_{\text{peak}} = 55^\circ$  at  $r/L_0 = 0.55$ . However, because of the presence of wing-like shape, compression at  $\theta \neq 0$  is a different shape-dependent feature compared to HIFiRE-5. The observation above could be concluded as near-field effect and hint future modeling efforts.



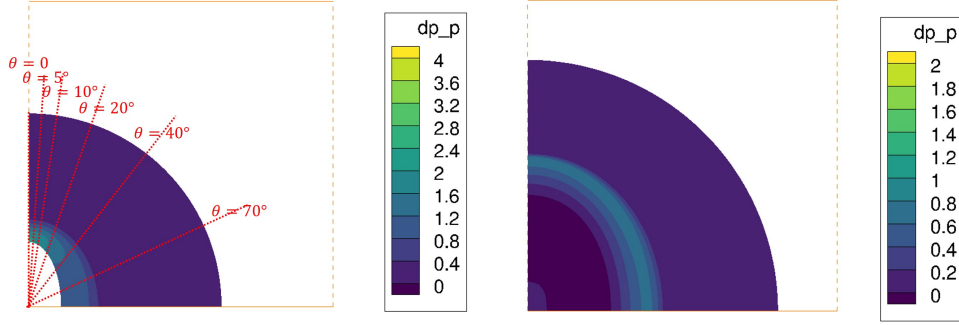
**Fig. 17** Lift  $L$ , drag  $D$ , and lift-to-drag ratio  $L/D$  as functions of angle of attack  $\alpha$  for a nosed elliptic cone (the first sector of HIFiRE-5) at  $M_\infty = 5$ .



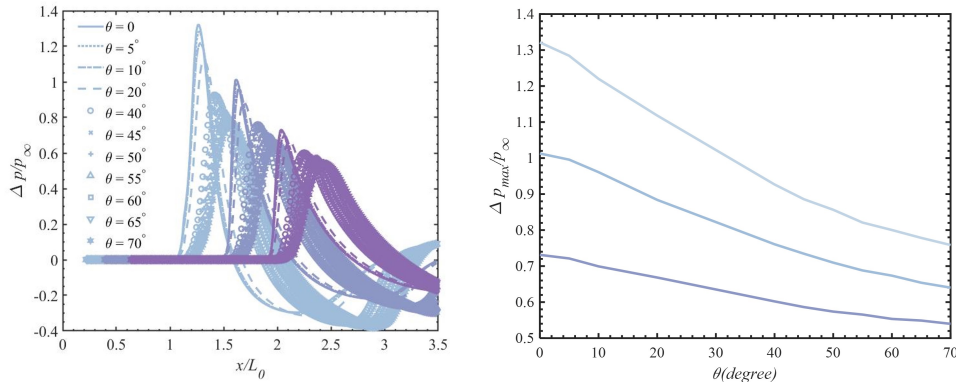
**Fig. 22**  $\theta$  dependency of normalized over-pressure  $\Delta p/p_\infty$  (left) and normalized maximum over-pressure  $\Delta p_{max}/p_\infty$  as a function of  $\theta$  (right) for a boost-glide vehicle (BGV) with a sting mount at zero angle of attack and  $M_\infty = 5$ :  $r/L_0 = 0.27$  (light blue),  $r/L_0 = 0.41$  (dark blue) and  $r/L_0 = 0.55$  (purple).

By extracting over-pressure signature at  $\theta = 0$  from the CFD data, our proposed line-distributed theory was also validated in Fig. 23. As can be seen from the figure, the drag corrections in Sec. II.C significantly improve the prediction.



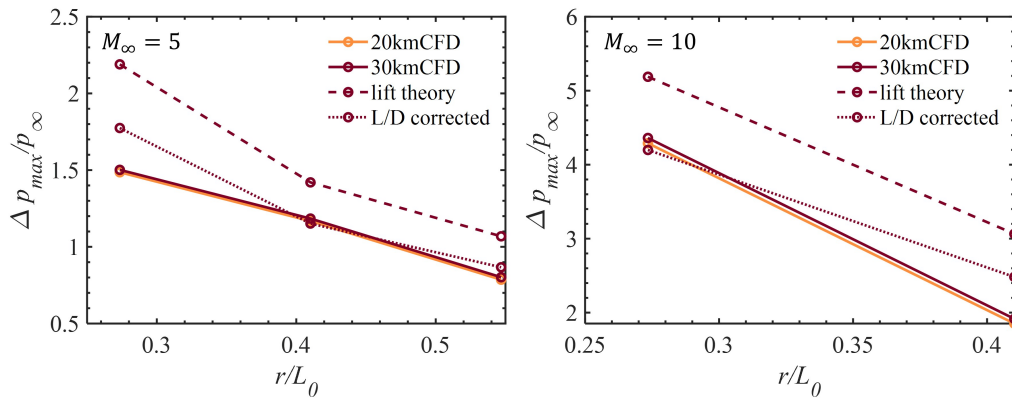


**Fig. 18** Slices of normalized over-pressure  $\Delta p/p_\infty$  contours at  $x/L_0 = 0.94$  (left) and  $x/L_0 = 1.71$  (right) of a nosed elliptic cone (the first sector of HIFiRE-5) at  $M_\infty = 5$ .



**Fig. 19** Normalized over-pressure sampled at different  $\theta$  for  $r/L_0 = 0.35$  (light blue),  $r/L_0 = 0.46$  (dark blue) and  $r/L_0 = 0.58$  (purple). Left:  $\Delta p/p_\infty$  as a function of  $x/L_0$  at different  $r/L_0$  locations and different azimuthal angles  $\theta$ , right:  $\Delta p_{max}/p_\infty$  as a function of azimuthal angle  $\theta$  at different  $r/L_0$  locations.

Compared to the HIFiRE-5 cases, the near-field prediction is much better in the BGV cases. The reasons of the unexpected excellence are (i) the relatively flat over-pressure vs.  $\theta$  in the neighbor of  $\theta = 0$  profile in the near field (see Fig. 22), and (ii) an excellent lift-to-drag ratio  $L/D$  prediction with very small error.



**Fig. 23** Normalized maximum over-pressure  $\Delta p_{max}/p_\infty$  a boost-glide vehicle (BGV) with a sting mount at zero angle of attack:  $M_\infty = 5$  (left),  $M_\infty = 10$  (right). The CFD data is extracted at  $\theta = 0$ . Solid lines are the CFD data, dashed lines are the line-distributed theory prediction without drag corrections, and dotted lines are the theory prediction with drag corrections.

### C. Mid-field and far-field over-pressure signatures: theory vs. NASA sBOOM

Following the standard CFD + 1D wave propagation code pipeline, the near-field data from CFD simulation is used as the input data (i.e., initial condition) for NASA sBOOM software [9] to propagate the over-pressure  $\Delta p$  signatures to middle and far fields, using the 1976 standard atmosphere model [34], for any given cruise altitude  $H$ . As indicated in Sec. III.B, the shock intensity induced by hypersonic flows in the near-field is  $\sim O(1) - O(10)$  and hence much stronger than a common value of  $\sim O(0.01)$  in the supersonic regime. For this reason, much larger space for the relaxation process (space occupied by expansion and peak decay) is needed. To get the full over-pressure signatures in middle and far fields, it requires about 80% – 99% zeros paddings (i.e., the partition of the mesh with 0 value surrounding the original signature at the beginning of the propagation process for wave propagation) in the sBOOM settings to avoid the over-pressure signatures partially escaping the computational domain in middle and far fields. In contrast, in the supersonic regime,  $\sim 30\% - 40\%$  zeros paddings are sufficient. The normalized maximum over-pressure  $\Delta p_{\max}/p_{\infty}$  generated by sBOOM is then used to validate the proposed line-distributed formulations in middle and far fields.

#### 1. Validation of line-distributed formulations by NASA sBOOM data and effect of stratified atmosphere

The classical point-source blast wave theory by Plooster [23] was derived from the shock propagation in homogeneous atmosphere (i.e. constant  $p_{\infty}$ ). As a result, the new line-distributed theory proposed in this paper also assumes homogeneous atmosphere propagation. However, the exact atmosphere during propagation process is stratified  $p_{\infty} = p_{\infty}(h)$  and  $h$  is the signature altitude, which is usually modeled by the 1976 standard atmosphere model [34]. The discrepancy makes the far-field  $\Delta p_{\max}/p_{\infty}$  be significantly over-estimated since the ambient pressure at cruise altitude  $H = 20\text{km}$  is only  $\sim 5500\text{Pa}$ , which is only 5% of the ambient pressure at the sea level (1 atm). Such discrepancy is also confirmed in Fig. 24, the prediction given by the theory only works for the near field and part of the middle field. When  $r/L_0$  exceeds a critical value, the sBOOM data starts to curve down and deviate from the theory prediction, as a result of stronger stratified atmosphere effect.

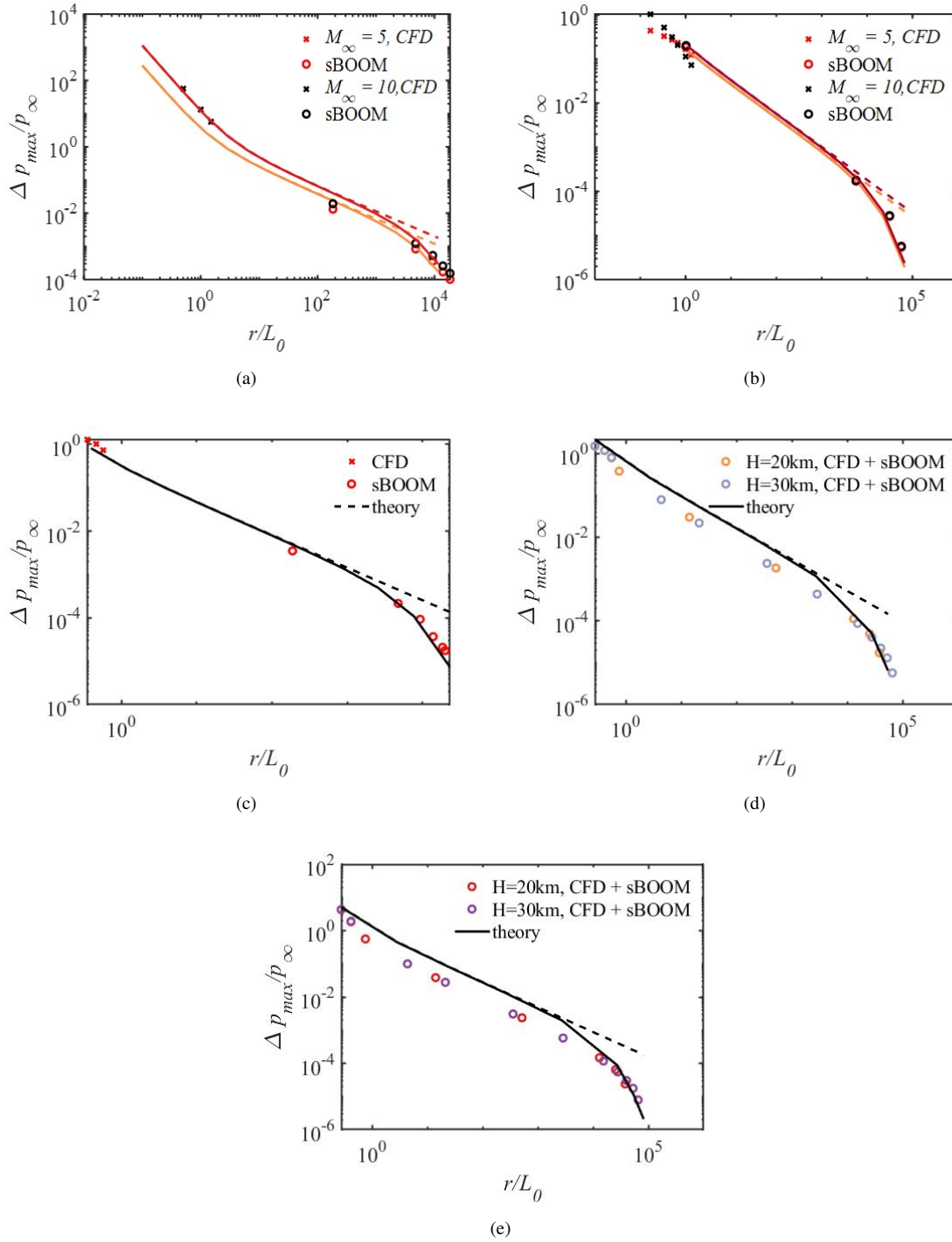
For better prediction accuracy, bridging  $\Delta p_{\max}/p_{\infty}$  in stratified atmosphere back to  $\Delta p_{\max}/p_{\infty}$  in homogeneous atmosphere should be derived starting from a ray-tracing theory. Here, with the simplest consideration of a  $0^{\text{th}}$ -order approximation, a correction coefficient  $C_{\text{amp}} = p_{\infty}(H)/p_{\infty}(h)$  (assuming negligible impact of the gradient of the ambient pressure) is multiplied to the homogeneous solution given by the theory, i.e.,

$$\left. \frac{\Delta p_{\max}}{p_{\infty}} \right|_{\text{stratified}} = C_{\text{amp}} \left. \frac{\Delta p_{\max}}{p_{\infty}} \right|_{\text{homogeneous}}. \quad (23)$$

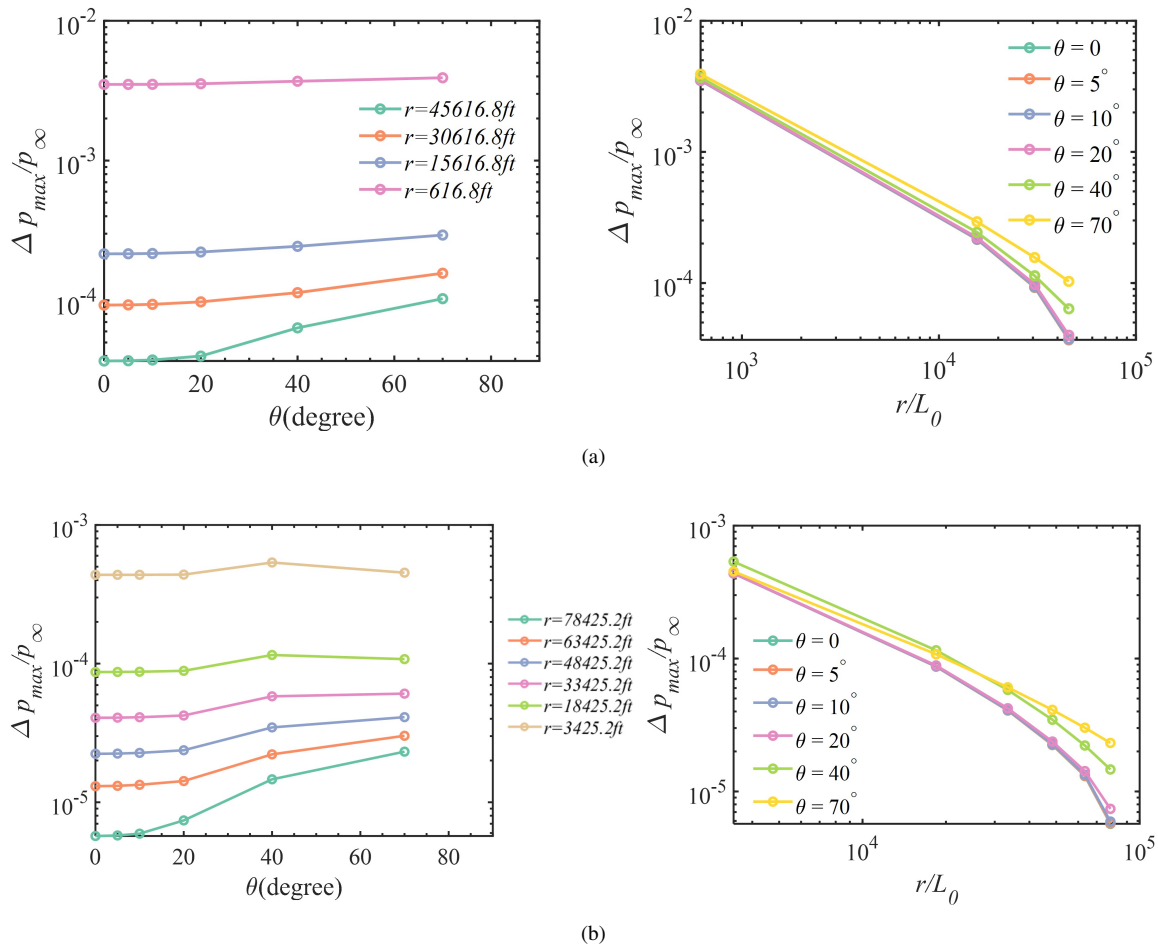
As shown in Fig. 24, even with such a simple far-field correction, the agreement of far-field theory prediction with the data given by sBOOM is improved significantly and matches nicely in the far-field for all the cases simulated in this work.

#### 2. Sonic boom carpet: effect of azimuthal angle $\theta$

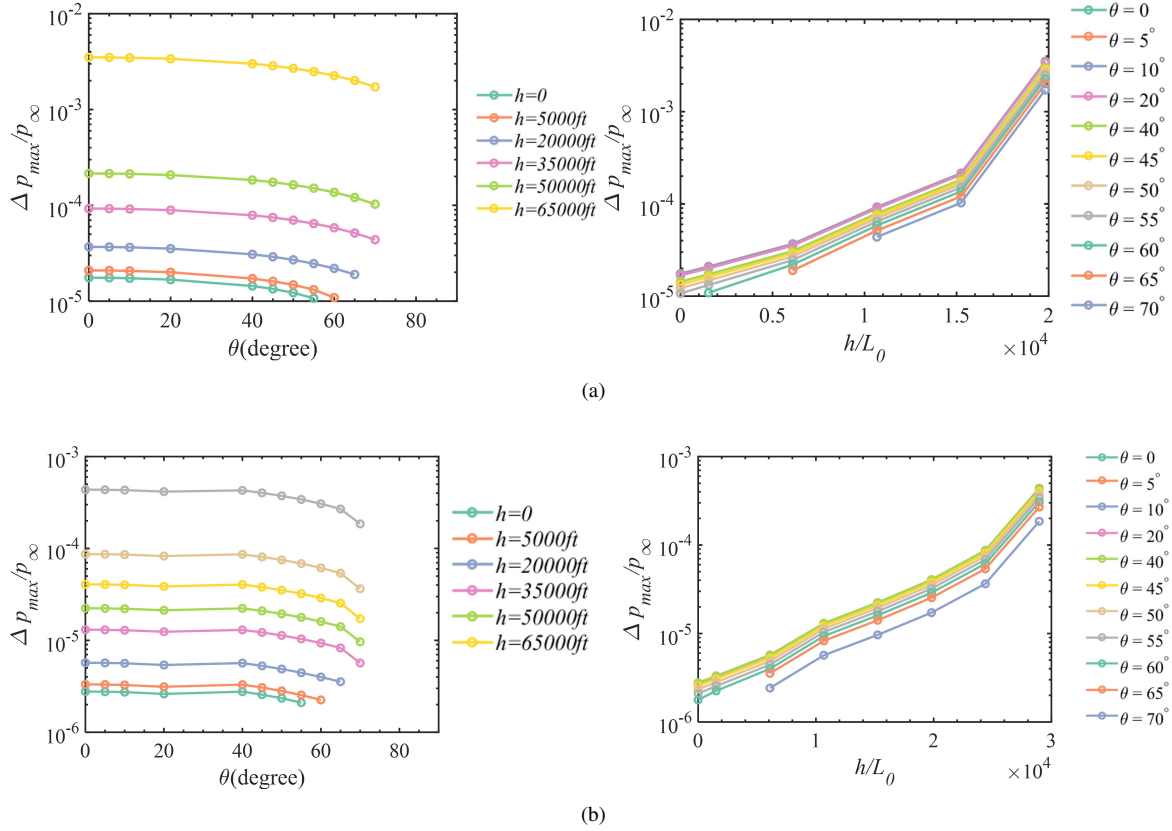
In both oHIFiRE-5 and BGV cases at  $\alpha = 0$ , it is verified that their near-field over-pressure could variate significantly with azimuthal angle  $\theta$ , but the variation will decay. Basing on this observation, for homogeneous atmosphere, it is expected that the  $\theta$  dependence would disappear asymptotically. Hence it is valuable in modeling perspective to examine the  $\theta$  dependence after far-field propagation. In this section, a HIFiRE-5 case starting propagation from a cruise altitude of  $H = 20\text{ km}$  and a BGV case starting propagation from a cruise altitude of  $H = 30\text{ km}$  are investigated. Due to the fact of stratified atmosphere, both radius  $r$  and altitude  $h$  are engaged in the far-field propagation process. Standing at the consistency to the near-field results, it is still more preferred to select  $r$  as the major basis. However, as plotted in Fig. 25, the far-field difference at fixed  $r$  is even growing asymptotically, which is opposite to the observation we made in the near field. This observation hints an alternative  $\theta$  effect in the propagation process.



**Fig. 24** Far-field comparison of the line-distributed theory with and without far-field correction for stratified atmosphere against sBOOM data for various geometries: (a) a sphere; (b) a nosed cone; (c) an elliptic cone (the first sector of HIFiRE-5); (d) a boost-glide vehicle (BGV) with a sting mount at  $M_\infty = 5$ ; and (e) the same BGV at  $M_\infty = 10$ . Solid lines indicate theory prediction with the far-field correction, and dashed lines indicates theory prediction without the far-field correction.



**Fig. 25** Carpet plot:  $r$ -base or at the cylindrical surface for (a)  $\theta$  dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $r$  (left) and radius dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $\theta$  (right) for HIFiRE-5; and (b)  $\theta$  dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $r$  (left) and radius dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $\theta$  (right) for BGV.



**Fig. 26 Carpet plot:  $h$ -base or at the cylindrical surface for (a)  $\theta$  dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $h$  (left) and radius dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $h$  (right) for HIFiRE-5; and (b)  $\theta$  dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $h$  (left) and radius dependency of  $\Delta p_{\max}/p_{\infty}$  at different  $h$  (right) for BGV.**

Alternative to  $r$ , we can also plot the far-field results under an altitude  $h$  basis shown in Fig. 26 (note that under same  $h$ ,  $r$  are not the same). The variation of  $\Delta p_{\max}/p_{\infty}$  at different  $\theta$  is then significantly diminished in the far-field.  $\Delta p_{\max}/p_{\infty}$  is found to be almost constant when  $|\theta| < 20^\circ$  during the whole propagation process. When  $\theta > \theta_c$  at large  $r$  or small  $h$  (or in another word, small  $h$ ), the ray tube is reflected by the atmosphere and hence the signature is truncated.

## IV. Conclusions

In this paper, a novel line-distributed energy impulse theory for hypersonic vehicle geometries with and without lift is developed. Due to the consideration of local lift distribution, it is capable to model the stronger shock below the vehicle compared to the classical point-source blast wave theory which can handle drag-dominant scenarios. Rapid and precise force prediction including lift and drag is achieved by a Newtonian aerodynamic algorithm with drag corrections. In the near field, due to the good predictions of both lift and drag, the theory gives a significant improvement of the over-pressure prediction with an error  $\sim 10\%$  in a wide spectrum of cases, including cone/HIFiRE-5 cases with  $\alpha \neq 0$  and boost-glide vehicle (BGV) with  $\alpha = 0$ . The theory works almost precisely in weak lift scenarios, but can also produce an acceptable prediction for strong lift scenarios. In the far field, a simple correction coefficient is employed to account for the stratified atmosphere effect. Incorporating this coefficient extends the capability of the current theory to model far-field propagation results with a good match for all scenarios tested in this study.

Azimuthal angle  $\theta$  dependence is studied in the simulation of HIFiRE-5 and BGV cases. Two  $\theta$  effects are identified: (1) In the near field,  $\theta$  effect is identified as a shape-induced spatial distribution which will decay when  $r$  is growing. (2) In the far field,  $\theta$  effect is identified as the deflection of sonic boom ray tube in the stratified atmosphere. Altitude-base analysis in the far field is more insightful compared with radius-base analysis, due to the significant role of stratified atmosphere. All of these findings will hint the future modeling of the  $\theta$  dependency of  $\Delta p_{\max}/p_{\infty}$ .

To conclude, a new line-distributed blast wave theory (considering the lift effect) for the rapid prediction of hypersonic sonic boom is developed. The theory is validated in both near field using CFD data and far field using wave propagation with an error  $\sim 10\%$  in various geometries, which shows significant improvement compared to the classical point-source blast wave theory, which can only handle drag-dominant scenarios. Azimuthal angle dependence is also investigated and two major effects are found, which could hint the future modeling efforts.

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